

1st Sem
Internals

Siliguri Institute of Technology
Department of Engineering Sciences and Humanities
1st Internal Examination – 2022 (Odd Semester)
Section-A, B, C & D

Paper Name: Mathematics IA
Full Marks: 25

Paper Code: BS-M 101
Time: 01 Hour

Group -A

1×5=5

(Answer any five Questions)

- Q1. (i) The value of the determinant $\begin{vmatrix} 100 & 101 & 102 \\ 105 & 106 & 107 \\ 110 & 111 & 112 \end{vmatrix}$ is [BS-M101.3]
(a) 2 (b) 0 (c) 405 (d) -1
- (ii) For any orthogonal matrix A, $\det A$ is equal to [BS-M101.3]
(a) 0 (b) 1 (c) ± 1 (d) -1
- (iii) The system of equations $x + 2y - z = 2$; $4x + 8y - 4z = 8$ has [BS-M101.3]
(a) infinitely many solution (b) no solution (c) a unique solution (d) none of these
- (iv) The matrix $\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ is [BS-M101.3]
(a) symmetric (b) skew-symmetric (c) singular (d) orthogonal
- (v) If λ be an eigen value of an $n \times n$ square matrix A then λ^{-1} is an eigen value of [BS-M101.3]
(a) A^2 (b) A/λ (c) A^T (d) A^{-1}
- (vi) In R^3 , the set of vectors $\{(1,0,1)(0,1,0)(0,0,1)\}$ will be [BS-M101.4]
(a) linearly dependent (b) linearly dependent & basis (c) linearly independent (d) none
- (vii) In R^3 the set $W = \{(x, 2y, 3z): x, y, z \in R\}$ is a [BS-M101.4]
(a) vector space (b) sub space (c) not a sub space (d) none of these

Group -B

4×5=20

(Answer any four Questions)

2. Find the rank of the matrix $\begin{bmatrix} 1 & 3 & 4 & 3 \\ 3 & 9 & 12 & 3 \\ 1 & 3 & 4 & 1 \end{bmatrix}$ [BS-M101.3]

3. For what value of λ and μ the following system of equations [BS-M101.3]

$$\begin{aligned} x + y + z &= 6 \\ x + 2y + 3z &= 10 \\ x + 2y + \lambda z &= \mu \end{aligned}$$

- has (i) no solution (ii) a unique solution (iii) an infinite number of solution

4. If $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ find all eigen values of A and obtain all the eigen vectors corresponding to the eigen values. [BS-M101.3]

5. Diagonalise the matrix $\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ and find the matrices which diagonalise it. [BS-M101.3]

6. Show that the set $W = \{(x, y, z) \in R^3: x - 3y + 4z = 0\}$ is a subspace of R^3 [BS-M101.4]

7. Find the basis and dimension of the subspace W of R^3 , where [BS-M101.4]

$$W = \{(x, y, z) \in R^3: x + 2y + 3z = 0\}$$

Section: B, C, D, E

Time Allotted: 1 Hour

(All the questions are aligned to CO1)

Full Marks: 25

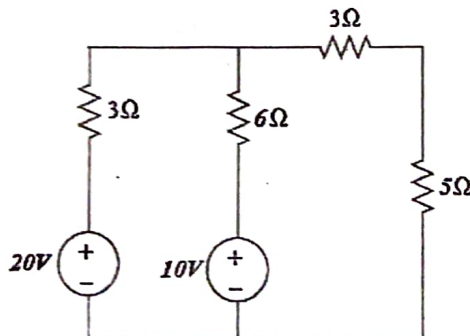
A. Answer Any Five Questions:

(1X5=05)

1. There are 4 resistors in a given series circuit and each resistor has 2.5V dropped across it. The source voltage (a) is 2.5V (b) depends on current value (c) depends on resistor value (d) 10V
2. For an ideal current source its internal resistance is (a) zero (b) infinite (c) none of these
3. The total resistance of a parallel circuit having two parallel resistors is $1\text{ k}\Omega$. One of the resistors is $2\text{ k}\Omega$. The other resistor value is (a) 0 (b) $2\text{ k}\Omega$ (c) $4\text{ k}\Omega$ (d) $8\text{ k}\Omega$
4. If R_{Th} be the Thevenin equivalent resistance & R_N be the Norton equivalent resistance of a network across the load terminals, then (a) $R_{Th} > R_N$ (b) $R_{Th} < R_N$ (c) $R_{Th} = R_N$ (d) None of these
5. The instantaneous AC voltage is expressed as $v = 141 \sin \theta$, the RMS value of voltage is (a) 141 V (b) 199 V (c) 100 V (d) 50 V
6. In an inductive circuit, Voltage _____ current. (a) Leads (b) Lags (c) is in phase with
7. An R-L series circuit has 10 ohm Resistance & 0.01 H Inductance. The circuit is supplied by an AC source with 100 V 50 Hz. The impedance of the circuit is given by _____. Ans 10.48 ohm

B. Answer Any Four Questions: (4X5=20)

8. Define Node, Branch, Loop & Mesh of a DC circuit with suitable diagram.
9. State & explain the KCL & KVL with suitable diagram.
10. State the Superposition theorem. What is its limitation?
11. Determine current through the 5Ω resistor using Thevenin's theorem.



12. An alternating current is given by $i = 100 \sin 188t$, determine (a) Frequency (b) RMS value of the current and (c) Peak factor
13. A $0.47\text{ }\mu\text{F}$ capacitor is connected to a 25 V, 200 Hz AC supply. Calculate the current flowing the circuit.

**MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL**

Paper Code : ESEE101 Basic Electrical Engineering

UPID : 001009

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- (i) What is fuse?
- (ii) What is the efficiency of power transfer when maximum transfer of power occurs?
- (iii) In a series R-L-C circuit, what is the condition for resonance?
- (iv) What is core type transformer?
- (v) What is basic principle of operation of Alternators / DC Generators?
- (vi) What is meant by duty-cycle?
- (vii) What is the function of a transformer?
- (viii) Define synchronous speed.
- (ix) What are the main classification of inverter?
- (x) What is secondary cell?
- (xi) What will be the impedance of a coil containing resistance of 8.0Ω and an Inductive reactance of 6.0Ω ?
- (xii) The full-load copper loss of a transformer is 1600 W. At half-load, what will be the copper loss?

Group-B (Short Answer Type Question)

Answer any three of the following

[5 x 3 = 15]

- 2. What is the significance of form factor in ac circuit? [5]
- 3. What are the salient features of distribution transformer? [5]
- 4. An AC current is expressed as $i = 14.14 \sin 314t$. Determine its a) RMS value, b) frequency, c) instantaneous value at $t = 2$ ms. [5]
- 5. What is meant by Regulation in a Transformer? [5]
- 6. Why are iron losses considered as constant losses in transformer? [5]

Group-C (Long Answer Type Question)

Answer any three of the following

[15 x 3 = 45]

- 7. (a) Explain the principle of series resonance. [5]
(b) What is half power frequency? [4]
(c) A coil of inductance 9 H and resistance 50Ω in series with a capacitor is supplied at constant voltage from a variable frequency source. If the maximum current of 1 A occurs at 75 Hz , find the frequency when the current is 0.5 A . [6]
- 8. (a) What is shell type transformer? Explain in detail. [6]
(b) The primary and secondary of a 25 kVA transformer has 500 and 40 turns, respectively. If the primary is connected to 3000 V , 50 Hz mains, calculate (i) Primary and secondary currents at full load; (ii) The secondary emf and (iii) The maximum flux in the core. Neglect magnetic leakage, resistance of the winding and the primary no-load current in relation to the full load current. [9]
- 9. (a) Justify that the average active power consumed by a purely inductive circuit is zero. [7]
(b) Explain the phasor diagram of RC series circuit with a neat circuit diagram. [8]

→ P.T.O

10. (a) A 100 kVA, 3300/200 volt, 50 Hz single phase transformer has 40 turns on the secondary, calculate:

(i) the values of primary and secondary currents.

(ii) the number of primary turns.

(iii) the maximum value of the flux.

If the transformer is to be used on a 25 Hz system, calculate

(iv) the primary voltage, assuming that the flux is increased by 10%.

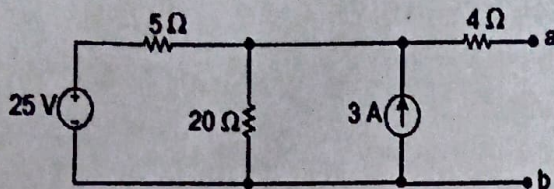
(v) the kVA rating of the transformer assuming the current density in the windings to be unaltered.

(b) Derive the emf equation of a single phase transformer. *half*

[6]

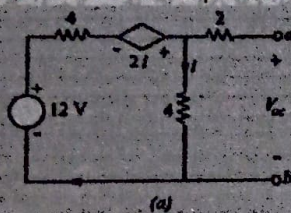
11. (a) Derive the Norton equivalent circuit of the given figure.

[8]



(b) Find Thevenin equivalent circuit for the following network.

[7]





Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 × 10 = 10]

- (i) Calculate the wavelength of an electron of energy 10 eV.
- (ii) Why do not we observe the de Broglie wavelength with a fast moving cricket ball?
- (iii) What is the phase difference between displacement and velocity of a particle executing SHM?
- (iv) Fill in the blank: Planck's formula for black body radiation can be derived from _____ statistics.
- (v) Why the Sound wave doesn't show the phenomenon of polarization?
- (vi) What is the expression of separation between interference fringes in double slit diffraction?
- (vii) A conductor of length L is placed in a magnetic field of magnetic flux density $\vec{B}$. If current I is flowing through the conductor then how much force will be experienced by it?
- (viii) What would be the amount of change in wavelength of a monochromatic X-ray photon, when it is scattered by carbon atoms at an angle 180° ?
- (ix) In which statistics, Pauli's exclusion principle is valid?

(x) Find $\vec{\nabla} \cdot \vec{r}$.

(xi) The differential equation of motion of a freely oscillating body is given by $\frac{d^2x}{dt^2} + 36\pi^2x = 0$. Find the natural frequency of the body.

(xii) At absolute zero temperature, what is the Fermi distribution of electrons when the energy of the system is greater than Fermi energy?

Group-B (Short Answer Type Question)

Answer any three of the following

[5 × 3 = 15]

2. What do you mean by harmonic motion? [5]
Give a graphical comparison among the following four types of harmonic motion:
i) simple harmonic motion,
ii) underdamped harmonic motion,
iii) overdamped harmonic motion and
iv) critically damped harmonic motion.
3. a) Deduce Brewster's law. [5]
b) Find the Brewster's angle of a glass slab of refractive angle 1.5. If the slab is immersed in water of refractive index $4/3$, find the angle of polarization.
4. What do you mean by irrotational vector? [5]
Show that $\vec{A} = (6xy + z^3)\hat{i} + (3x^2 - z)\hat{j} + (3xz^2 - y)\hat{k}$ is irrotational.
5. Write down the expression for the intensity of light due to Fraunhofer diffraction in a single slit and hence find the conditions for maxima and minima in the diffraction pattern. [5]
6. Evaluate $\vec{\nabla} \cdot \left(\frac{\vec{r}}{r^3} \right)$. [5]

Group-C (Long Answer Type Question)

Answer any three of the following

[15 × 3 = 45]

7. (a) What are positive and negative crystals? Give examples. What is optic axis of a crystal? [1+1+1+1]
(b) How can you distinguish between circularly polarized light and unpolarized light with help of quarter wave plate and Nicol prism. [4]

- (c) Explain why light wave can be polarized but sound wave cannot be polarized. [2]
- (d) Find the state of the polarization when x and y components of the electric fields are given by $E_x = E_0 \sin(wt + kz)$ and $E_y = E_0 \cos(wt + kz)$. [2]
- (e) A plane polarized light of wavelength 600 nm changes to circularly polarized light on passing through a quartz crystal cut parallel to optic axis. Calculate the minimum thickness to produce such effect. Given ($\mu_e - \mu_o$) = 0.005. [3]
8. (a) Define Magnetic susceptibility and permeability. Obtain the relation between them. [2+3]
- (b) Write the differences between diamagnetic, paramagnetic and ferromagnetic substances. [4]
- (c) What is hysteresis? Draw a hysteresis curve for a sample of iron. Explain the curve step by step. [1+2+3]
9. (a) What are dielectrics? Differentiate between polar and non-polar dielectrics with examples. [4]
- (b) Find the relation between electronic polarizability and atomic radius. [4]
- (c) Establish the relation $\vec{D} = \epsilon_0 \vec{E} + \vec{P}$, where the parameters have their usual meaning. [3]
- (d) A potential of 10V is applied across a parallel plate capacitor having area of $6.45 \times 10^{-4} m^2$ and plate separation is 0.002m. If a material having dielectric constant of 6.0 is placed within the region between the plates calculate the amount of charge stored in the plate and the net amount of polarization. [4]
10. (a) Write down the expression for intensity distribution of single slit diffraction and explain the symbol used. [2]
- (b) Sketch how the intensity varies with angle of diffraction. [2]
- (c) Find the values of the angle at which minima are located. [2]
- (d) Find the condition that the angle satisfies in order to be the position of a secondary maximum. [2]
- (e) Describe the fringe pattern when white light is incident on a plane transmission grating. [3]
- (f) A light of wavelength 6×10^{-5} cm falls normally on a slit of width 0.2 mm. Calculate i) the total angular width of the central maximum ii) the linear width of central maximum on a screen placed 2 m away. [4]
11. (a) Write down Maxwell's wave equations for free space with their physical significance. Hence prove that velocity of EM wave is equal to the velocity of light. [5]
- (b) Write down the Maxwell's modification of Ampere's law. [2]
- An electric wave is represented by $E = E_0 \cos(1000x - 5000t)$. Calculate the wavelength and speed of propagation of the wave. Show that the displacement current is negligible compared to the conduction current at any frequency when the conductor is subjected to an alternating field. [2+2]
- (c) A circular conducting disc of 5 cm radius maintains an angular motion of 1000 rpm with its plane perpendicular to the magnetic field. If due to this motion, the induced e. m. f. between the center and the edge of the disc is 3.22 mV, calculate the magnetic flux density. [4]



MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : BSM101 Mathematics -IA

UPID : 001004

Full Marks : 70

Time Allotted : 3 Hours

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

[1 x 10 = 10]

1. Answer any ten of the following :

(i) Solve for x and y using CRAMER'S RULE ✓
 $3x + 2y = 16$
 $7x - y = 19$

(ii) What are the necessary condition for the maclaurin's expansion to be true for function f(x) is

(iii) If $f(x) = 2|x| + |x-2|$ then find the value of $f(1)$

(iv) If $y = 10^{5x}$ then find the value of y_n

(v) Calculate the value of

$$\int_0^{\infty} e^{-x} dx$$

(vi) What are the coefficients of the equation obtained during the elimination called

(vii) For which value of x will $(x-1)(3-x)$ have its maximum $x=2$

(viii) If A is matrix of order 3×5 then rank of A^T is at most 5×3

(ix) Find the Wronskian of $1, e^x, e^{2x}$

(x) If p1 be an eigen value of the matrix A then what is the eigen value of the matrix $A+1$ $p_1 + 1$

(xi) If $A^2 - A + I = 0$, then $A^{-1} = \frac{A - I}{A^2 + I}$

(xii) If the eigen values of the matrix

$$\begin{pmatrix} 2 & 3 \\ x & y \end{pmatrix}$$

Group-B (Short Answer Type Question)

Answer any three of the following

[5 x 3 = 15]

2. Prove that set of all second order real square matrices is a vector space with respect to addition and multiplication of a matrix by a real number [5]

3. Prove that the Eigen values of a real symmetric matrix are all real [5]

4. Find the value of [5]

$$\int_0^{\pi/2} \sin^3 \theta \cos^3 \theta d\theta = \frac{16}{60}$$

5. Show that the maximum value of $x+1/x$ is less than its minimum value [5]

6. State Cayley Hamilton theorem and verify for matrix [5]

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Group-C (Long Answer Type Question)

Answer any three of the following

[15 x 3 = 45]

7. (a) Verify Rolle's theorem for the function $f(x) = e^{-x} \sin x$ on $[0, \pi]$. [5]

(b) Using MVT prove that

$$0 < \frac{1}{x} \log \frac{e^x - 1}{x} < 1.$$

[5]

(c) Using MVT prove that

$$\sin 46^\circ \sim \frac{1}{2} \sqrt{2} \left(1 + \frac{\pi}{180} \right).$$

[5]

8. (a) Examine the consistency of the following system of equations and solve $2x+2y-4z=8, 2x+3y+2z=8, x+y-z=7/2$

[5]

(b) Find the rank of

[5]

$$\begin{pmatrix} 0 & 0 & 5 & -3 \\ 2 & 4 & 3 & 5 \\ -1 & -2 & 6 & -7 \end{pmatrix}.$$

(c) If A is skew symmetric matrix then prove that $(I-A)(I+A)^{-1}$ is orthogonal matrix

[5]

9. (a) Using definition of beta function prove that

[5]

$$\int_0^{\pi/2} \cos^4 x dx = 3\pi/16$$

(b) Evaluate

[5]

$$\int_0^\infty e^{-x^2} dx$$

(c) show that

[5]

$$\int_0^1 x^{3/2} (1-x)^{3/2} dx = 3\pi/128$$

10. (a) Using Lagrange's MVT find the approximate value of

[5]

$$\sqrt[5]{65}$$

(b) Using MVT prove that

[5]

$$x < \sin^{-1} x < \frac{x}{\sqrt{1-x^2}}, \text{ if } 0 < x < 1.$$

(c) Expand the function $f(x) = \sin x$ in powers of x with Lagrange's form of remainder

[5]

11. (a) Find the value of

[5]

$$\lim_{x \rightarrow 0} x^{2 \sin x}$$

(b) Find the value of

[5]

$$\lim_{x \rightarrow 0} \cot x^{\sin x}$$

(c) Find the value of

[5]

$$\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{1}{x^2}} = \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{1}{x^2}}$$